Methods
Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- Acoustic matrices: the phone token's MFCCs (30 coefficients; 10 frames)
- Articulatory matrices (EMA/annotated MRI):
  - The temporal dimension of the video clip for each token was normalized to 5 frames (AE-MRI-annotated) & 10 frames (AE-EMA).
- MFCCs/x, y, (z) **coordinates** → 1D CNN + temporal pooling + MLP → <u>1024D latent matrix</u>

Methods

Distinctiveness

- In each bigram, phone2's **distinctiveness** w/ phone1 was calculated as the mean Euclidean distance between their **articulatory/acoustic matrices**.
- Articulatory matrices (ultrasound/MRI):
  - The temporal dimension of the video clip for each token was normalized to 5 frames (FR-MRI/AE-MRI) & 10 frames (BE-US).
  - Raw video → 3D CNN + MLP → <u>1024D latent matrix</u>